

SPECTROPHOTOMETER ON-THE-GO: THE DEVELOPMENT OF A 2-IN-1 UV-VIS PORTABLE ARDUINO-BASED SPECTROPHOTOMETER

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ABSTRACT

The spectrophotometer is an important common analytical equipment that measures the concentration of analytes via their absorbance of light at specific wavelengths. However, many laboratories with limited resources cannot afford such equipment, and the lack of portability and requirement of constant power supply prevents accurate measurements in time-sensitive field tests. Here we describe the development of a portable and low-cost Arduino-based 2-in-1 UV-Vis spectrophotometer built from off-the-shelf electronics and 3D printing. The assembled device has an overall dimension of 105 x 90 x 140 mm and has photodiodes capable of detecting light wavelengths at a range of 190 – 1100 nm, sending the results to its corresponding SpectBT UV-Vis Pro smartphone Android app via Bluetooth®. The device functions were compared with a commercial research grade spectrophotometer with respect to measuring nucleic acid and protein biomolecules to determine the limit of detection and sensitivity. It was shown that our assembled spectrophotometer – ‘APD UV-Vis SpectBT’ - achieved better performance at ~595 nm measurements while retaining comparable performance for 280 nm UV measurements. These findings demonstrate that certain laboratory equipment could be assembled to meet the needs of various laboratories.

Keywords: UV-Vis, Arduino, Spectrophotometer, Scientific Phone Apps, Absorbance, 3D Printing

1. Introduction

The ultraviolet-visible light (UV-Vis) spectrophotometer is an analytical tool that measures analyte absorbances of light at certain wavelengths [1]. It is widely used for purity measurements in the environment [2], assessing F&B (food & beverage) products [3], and quantification of biomolecules in biomedical/clinical uses [4].

Colours have defined wavelengths within the visible light spectrum, and these are absorbed or reflected uniquely by different matter. It is this property of unique absorption and reflection that the UV-Vis spectrophotometer measures to quantify substances [5].

Along with the relative high costs of commercial spectrophotometers, additional software and computers for data collation and analysis are required for spectrophotometric analysis. Such requirements are obstacles to less well-off laboratories, hindering adoption. Even when costs are not intrinsic barriers, the stationary nature of such a device restricts spectrophotometric measurements predominantly to the laboratory environment [6], disallowing immediate analysis of time sensitive experiments in the field or place of sampling.

Many field tests in environmental monitoring such as the testing of water quality [7], off-site clinical screening of body fluids such as the detection of blood [8] or protein in urine [9] are time sensitive, requiring rapid transport in low temperatures to laboratory test sites for accurate measurements. For optimal accuracy, such processes can benefit from immediate on-site analysis using portable spectrophotometers before confounding variables such as sample degradation or microbial growth set in to affect analytical outcomes.

The need for portable laboratory equipment and self-assembly towards spatial geographical freedom of access is also evident in the recent pandemic measures where access to such analytical equipment is restricted from lockdown measures [10], bringing research dependent on these equipment to a halt with cascading repercussions beyond scientific

progress and scientific career progressions. These effects have thus spurred a movement towards DIY (do it yourself) assembly options in scientific research.

Commercial manufacturing of the essential spectrophotometer casings is typically manufactured by injection or blow moulding that generate high amounts of waste material and are prohibitively expensive (due to mould production). With recent advancements in technology, prototyping and small-scale production can now utilize 3D printing, producing less waste and shortening the manufacturing lead time for quicker development of products.

Leveraging on the recent smartphone revolution comprising of scientific-use apps [11], [12], [13] that allow on-the-go lab analyses [14], [15], [16], [17] and facilitate field analysis [18] outside the lab environment, we employ the same approach by further developing our previous visible-light spectrophotometer [19] towards a portable and low-cost 2-in-1 UV-Vis spectrophotometer coupled with its own dedicated mobile app for on-the-go spectrophotometric analysis in both the UV and visible (Vis) light range that is comparable to commercial UV-Vis spectrophotometers.

2. Materials and Methods

2.1. Components and Specifications

Arduino Nano (ATmega328P), dual operational-amplifier (op-amp) TL072CP (Texas Instruments), OSD5.8-7Q Full Spectrum Si Photodiode (Centronic) and an AT-09 BLE Bluetooth module were used to assemble the device. The circuit runs on a 7.4V Lithium Polymer (LiPo) battery (Limskey) and the TL072CP op-amp amplifies the small signal voltage from each photodiode to a usable signal voltage. A 3-way latching switch toggles light emittance from either the 265 – 310 nm UV LED (Houkem) which has a peak emission from 275 – 285 nm, or the 4500K white light LED (Cree Inc.). The transmittance of each light source is detected by the two OSD5.8-7Q Photodiodes (with a detection range of 190 – 1100 nm) – one for UV and one for visible light, with both converting the light absorbed into

electrical current. The AT-09 BLE Bluetooth module (Sincere & Promise) is incorporated for Bluetooth transmission of data to our smartphone Android app named APD SpectBT UV-Vis Pro.

Arduino was used for this endeavour for its ease in programming and low cost. While Raspberry Pi [20] is often utilized for multitasking operations, its cost and purpose were considered excessive for the project's requirement. Raspberry Pi was also reported to have high sensitivity to overvoltage when battery-powered [20], [21] whereas Arduino can easily rely on rechargeable batteries [22].

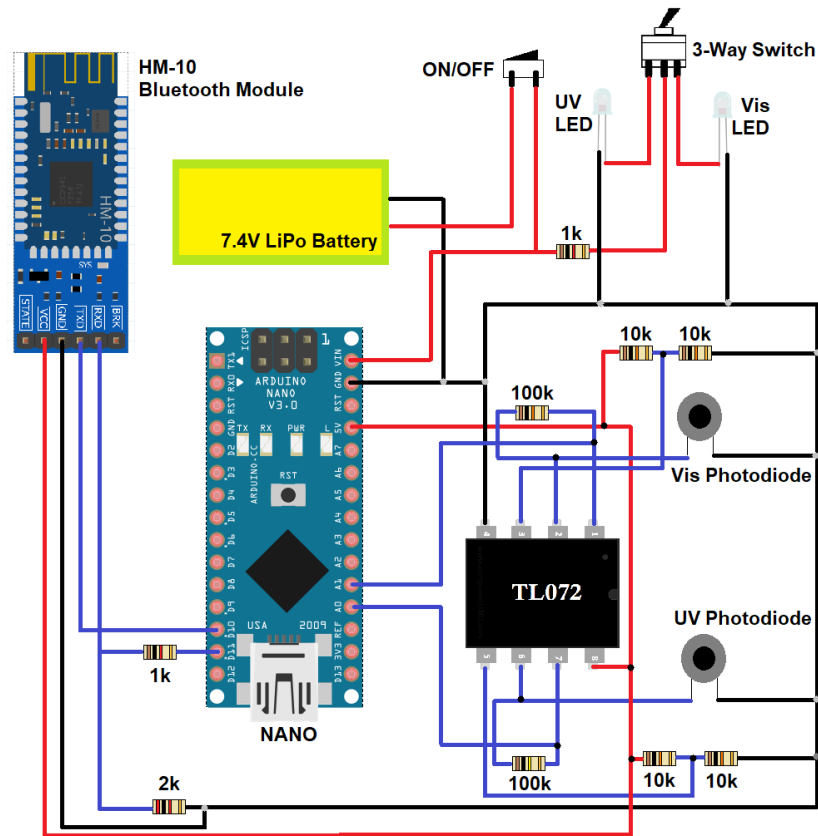


Figure 1. Schematic diagram of spectrophotometer

2.2. Physical Assembly and Design

The 3D printed casing and components of our APD UV-Vis SpectBT spectrophotometer, designed using Autodesk Tinkercad software [23], are of four main category of parts: the main body (Supplementary Figure 1), cuvette holder, the holder bolts and slide cover (Supplementary Figure 2). ~300g of black polylactic acid (PLA) filament (3D

Aura) was used, with the black colour selected for its low light reflectivity properties required in spectrophotometer interiors. The main body houses the electronic components such as the PCB (printed circuit board) and photodiodes. The two cuvette holders (Supplementary Figure 2) exteriors were imprinted with “P” and “Q” to indicate their respective 10 mm x 10 mm plastic (LP Italiana) and quartz cuvette (Dynamica) insertion slots. 3D printed bolts were used to hold the cuvette holders in place as well as rotation for the insertion and removal of cuvettes.

Figure 2 and a video (<https://youtu.be/6vz-fRfKJU8>) show the fully assembled spectrophotometer. The sliding cover ensures a light-tight environment while allowing access to internal components without having to dismantle the entire setup during the replacement or adjustments of parts (e.g. the battery). The ventilation holes at the lateral bottom half of the case provide help prevent overheating (Figure 2A) and a socket opening assists in direct charging of the LiPo battery by an IEC C7 connector which can supply about 24 hrs of continuous operation.

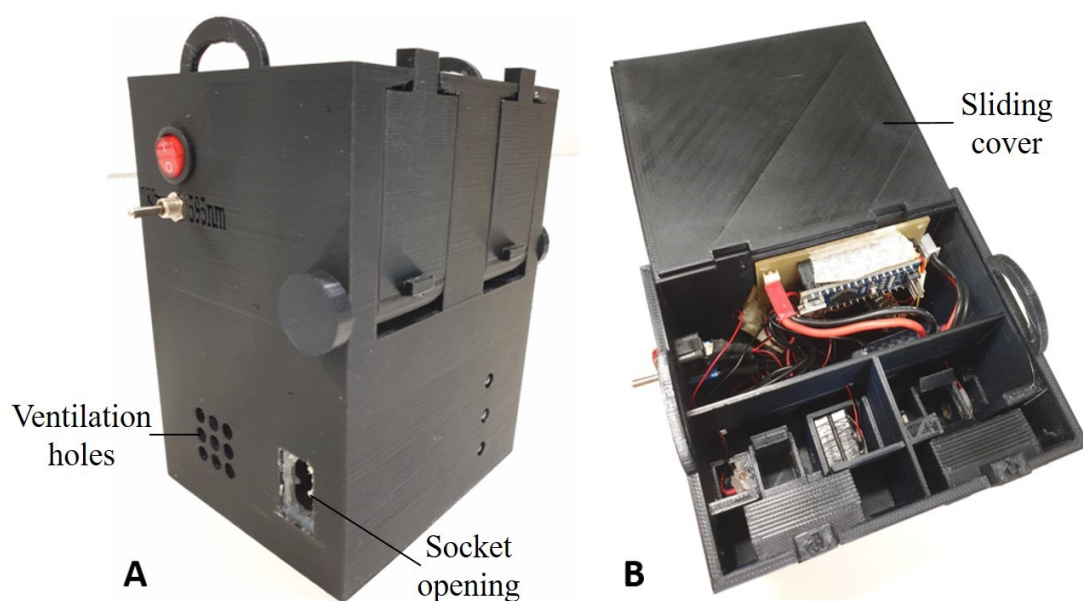


Figure 2. Overview of (A) APD UV-Vis SpectBT and (B) its internal components.

APD UV-Vis SpectBT is smaller and lighter than most commercial benchtop spectrophotometers (Supplementary Figure 3) and is battery operated. The total cost of

assembling the device (Supplementary Table 1) at ~S\$290 is ~14 times cheaper than the selling price of typical commercial spectrophotometers (Supplementary Figure 4). The advantage of our device over other DIY spectrophotometers is that it can measure in both UV-Vis spectrums whilst retaining compatibility of existing generic cuvettes (Supplementary Figure 4).

One indirect limitation of our device compared to some commercial nanospectrophotometers is the large sample volume of analyte (1.0 – 1.5 mL) required in the generic cuvettes to ensure compatibility with lab spectrophotometers. This results in limitations of biomolecules (e.g. DNA or protein samples) measurement where the samples are often limited. Yet, this can be easily addressed by performing dilutions. Forgoing compatibility with existing lab spectrophotometers, future advancements towards Nanophotometer technology [24] can be employed by applying surface tension properties of fluid to create a liquid column for spectrophotometric analysis (Supplementary Figure 5) [25].

2.3. Spectrophotometer Programming Arrangement

Arduino IDE 1.8.13 software [26] was used to program the system operation logic flow (Supplementary Figure 6) of the Arduino Nano. Transmittance signals from both UV and Vis photodiodes were processed by the Arduino Nano where the analog signals were converted to digital signals and transmitted to the APD SpectBT UV-Vis Pro app via the BLE Bluetooth® transmitter. This iteration was repeated after a 0.2 sec delay.

2.4. Mobile Application Programming Arrangement

The SpectBT UV-Vis Pro mobile app was developed using Android Studio IDE [27]. A flowchart of the internal processes in generating output data using the IDE as illustrated in Supplementary Figure 7. Initial calibration was performed by recording Abs_{Air} with no cuvette inserted. Abs_{Blank} was determined by implementing the Beer-Lambert equation (Equation 3) for the cuvette containing the ‘blank’ solution. Likewise, an initial ‘Sample Abs’

was also obtained for the cuvette containing the sample of interest. An output of the final Abs_{Sample} was established by subtracting Abs_{Blank} from Abs_{Air} . The final Abs_{Sample} readings are stored, deleted, or exported locally through APD SpectBT UV-Vis Pro app.

The UI (user interface) of SpectBT UV-VIS Pro app allows for simple navigation and quick access to the measured data. After calibration, Abs_{Sample} readings can be stored by tapping on the “Record” button. Users can view recorded data, re-calibrate (when changing reference solution) or switch light channels (Figure 3A). The data is stored in table format as a CSV file format (Figure 3B) by our StanXY app [28] for further plotting and quantification against a standard curve.

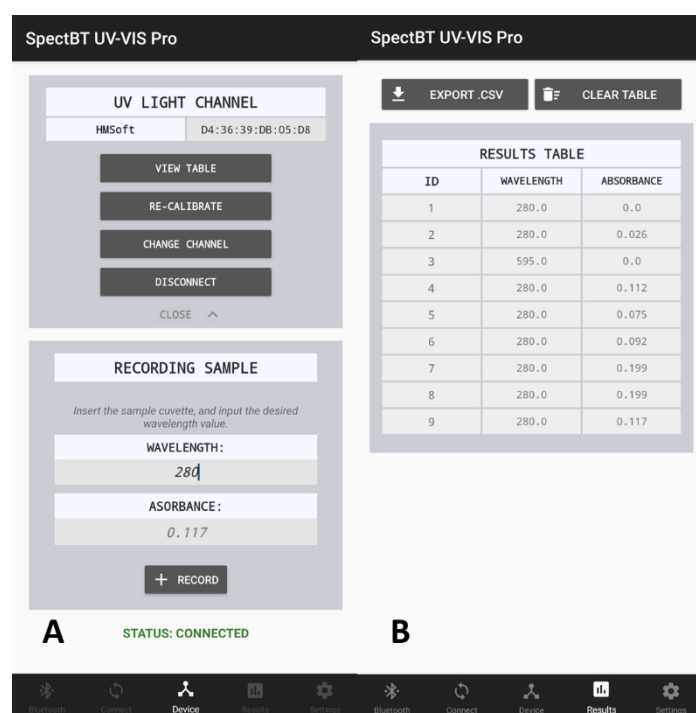


Figure 3. SpectBT UV-VIS Pro screenshots on (A) “Device” tab and (B) “Results” tab

The mobile app reduces the need for computers and additional display modules which would make APD UV-Vis SpectBT device larger. The combined used with the ubiquitous smartphone thus allows our device to be kept small, portable, low cost and affordable. The data in the app stored locally could also be shared via instant messaging apps or uploaded to Cloud-based storage (e.g. Google Drive, Dropbox, etc.). For future development, we aim to extend the apps and connectivity to iOS devices.

2.5. Operation of Spectrophotometer

A video demonstration of the device and app in operation could be found here: <https://youtu.be/hxRTLFIY-AA>. Depending on the required analysis, the user can toggle the latching switch to either UV at A_{280} or Vis at A_{595} , also reflecting the recording on the mobile app to the corresponding UV/Vis wavelengths. Quartz and plastic cuvettes were used for spectrophotometric analyses of A_{280} or A_{595} samples respectively. Analyses consisting of three main steps: 1) measurement of Abs in air (when no cuvette is inserted), 2) measurement of Abs of the blank solution (reference solution which contains little or no analyte of interest) and 3) measurement of Abs in the sample of interest were carried out for each operation. A minimum of 1.5 mL of the sample solution was dispensed into the cuvette, with the optically clear surfaces of the cuvette facing the light source and detector (Figure 4B). The sliding cover and cuvette holder were completely closed to ensure a light sealed environment.

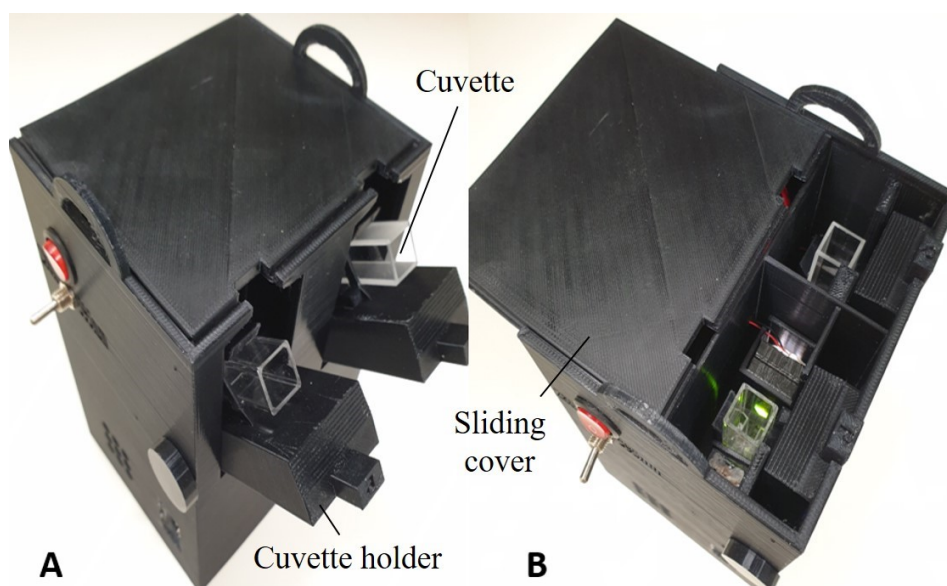


Figure 4. (A) Loading of cuvettes and (B) operation of Vis at A_{595} on assembled device

2.6. Experimental Setup

Nucleic acids and proteins absorb light at UV wavelengths, each with their respective peak Abs at 260 nm and 280 nm [29], [30]. The presence of aromatic tryptophan and tyrosine

allow for direct specific UV wavelength absorption without pre-processing with special reagents [30].

Since tyrosine and tryptophan composition numbers can vary greatly in different various proteins, colorimetric procedures (such as Bradford assay) are thus employed to get a more accurate estimate of protein concentration despite being less direct than A_{280} measurements. Assays like Bradford involve colorimetric reactions proportional to protein content where the dye forms a stabilized blue complex that absorbs light at 595 nm [31] with proteins through a series of electrostatic and hydrophobic interactions. The intensity of coloured dye is thus linearly correlated with the concentration of protein. By establishing a standard calibration curve using known protein concentrations of typically bovine serum albumin (BSA), the concentration of the protein in unknown samples can be derived from the linear sections via the Abs.

To validate the functionality of APD UV-Vis SpectBT, protein samples were quantified and compared with a commercial spectrophotometer (Implen NanoPhotometer P330) at A_{280} and A_{595} . 1.5 mL and 1.0 mL of samples were dispensed into a 10 x 10 mm quartz (A_{280}) and polystyrene (A_{595}) 10 mm path-length cuvette, respectively. All readings were taken in triplicates. Linear regression lines were expressed with the coefficient of determination R^2 to measure the goodness-of-fit.

For protein sample preparation, a series of BSA, Cat: A7906 (Sigma Aldrich) solutions ranging from 0.1 – 75 mg/mL were prepared by direct dilution from stock solution using deionized water. Protein samples were analysed using two methods: direct UV absorbance and Bradford assay. Using the UV channel, samples ranging from 0.1 – 40 mg/mL were measured directly at A_{280} . For the visible spectrum using the Bradford/Coomassie Protein Assay Kit, Cat: 23200 (Thermo Scientific), a concentration range of 0.1 – 10 mg/mL BSA (working range of assay is limited to 0.1 – 1.5 mg/mL) was

used. 30 µL of each BSA samples were added to 1.5 mL the Bradford/Coomassie reagent, mixed and incubated for 10 minutes prior to A₅₉₅ analysis.

The spectrophotometer was also validated using colorimetric analysis of Loop-Mediated Isothermal Amplification (LAMP) products for RNA detection. LAMP products (kind gift of Wei-Leong Chew, A*STAR, Singapore) were pooled based on three distinct coloured reaction solutions: pink, orange and yellow. These three solutions were indicative of the absence, ambiguity, and presence of target RNA in samples, respectively. The pooled samples were subjected to A₅₉₅ analysis. All readings were obtained in quintuplicates.

2.7. Measuring Principles

Quantification of sample molecules are derived from the Beer-Lambert's law within the linear range of respective standard curves.

Transmittance is the fraction of light intensity emitted through the sample and is governed by:

$$\text{Transmittance } (T) = \frac{I}{I_0} = 10^{-\epsilon cl} \quad (1)$$

While absorbance has the inverse logarithmic relationship to transmittance:

$$\text{Absorbance } (A) = -\log T \quad (2)$$

The combination of both Equation 1 and 2 derives a linear expression, also known as Beer-Lambert's equation:

$$A = -\log \frac{I}{I_0} = -\log \frac{I}{I_0} = \epsilon cl \quad (3)$$

Where I_0 and I are the intensities of incident and transmitted light, respectively. ϵ is the molar absorption coefficient in M⁻¹cm⁻¹, and is the sample dependent constant which determines the absorption of a particular wavelength by the sample. c is the molar concentration of sample in M, and l is the optical path length of sample which is typically 1

cm (Atkins & Paula, 2011). To comply with the Beer-Lambert Law, the absorbance unit (A) obtained from the measurements must fall within linear range of the spectrophotometer, where Abs is linearly proportional to the concentration.

3. Results and Discussion

A_{280} and A_{595} were obtained using our APD UV-Vis SpectBT and calibration curves using StanXY [28] were established to compare the performance against a commercial spectrophotometer (Implen P330).

3.1. Absorbance of Protein at 280 nm

Standard curves of BSA concentrations at 0.1 - 75 mg/mL on the Implen P330 and APD UV-Vis SpectBT showed the linearity of both calibration curves to be comparable. The Implen P330 has an Abs measuring range of 0.1 to 5.0 mg/mL while APD UV-Vis SpectBT has a higher measuring range of 3.1 to 75 mg/mL (Figure 5 and Supplementary Table 3). The APD UV-Vis SpectBT obtained a R^2 value of 0.9297 that is marginally lower than Implen P330 at 0.9959. While the APD UV-Vis SpectBT also demonstrated poorer repeatability with a high SE values (~ 0.042), it could measure up to 75 mg/mL of protein.

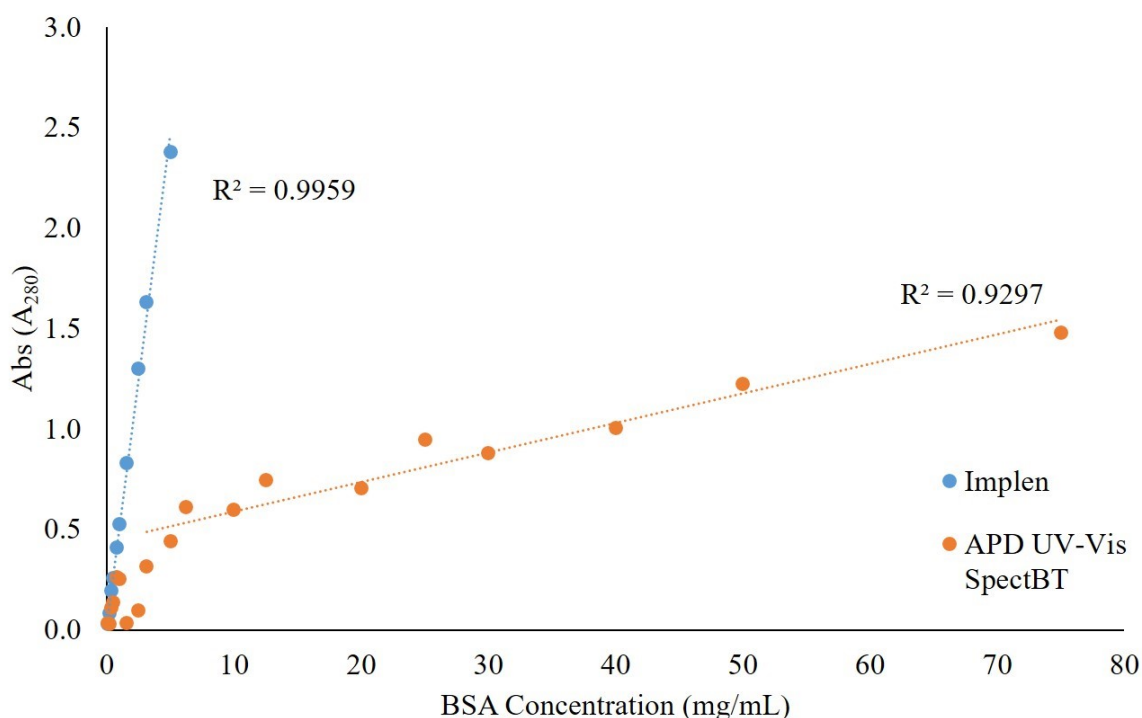


Figure 5. A_{280} standard curves of BSA from Implen P330 and APD UV-Vis SpectBT

The differences in absorption profiles between the devices are likely from the different light sources, detector sensors and internal algorithms. The Implen P330 employs a more advanced sensor capable with higher sensitivity compared with APD UV-Vis SpectBT use of off-the-shelf emitter and detector.

3.2. Absorbance of Protein at 595 nm

The A_{595} standard curves of the respective devices also showed comparable linearity (Figure 6). Unlike the 280 nm analysis, APD UV-Vis SpectBT ($R^2 = 0.9462$) had better R^2 readings than the Implen P330 ($R^2 = 0.906$) with similar gradients indicating comparable measurement sensitivities. While the recommended measuring range according to Bradford assay is defined to be within 0.1 - 1.6 mg/mL as observed in the Implen P330 (Supplementary Table 4), APD UV-Vis SpectBT can measure up to 3.1 mg/mL. This indicates a higher limit of detection for APD UV-Vis SpectBT for the Bradford assay, which also showed promising repeatability ($SE = 0$) at the various BSA concentrations.

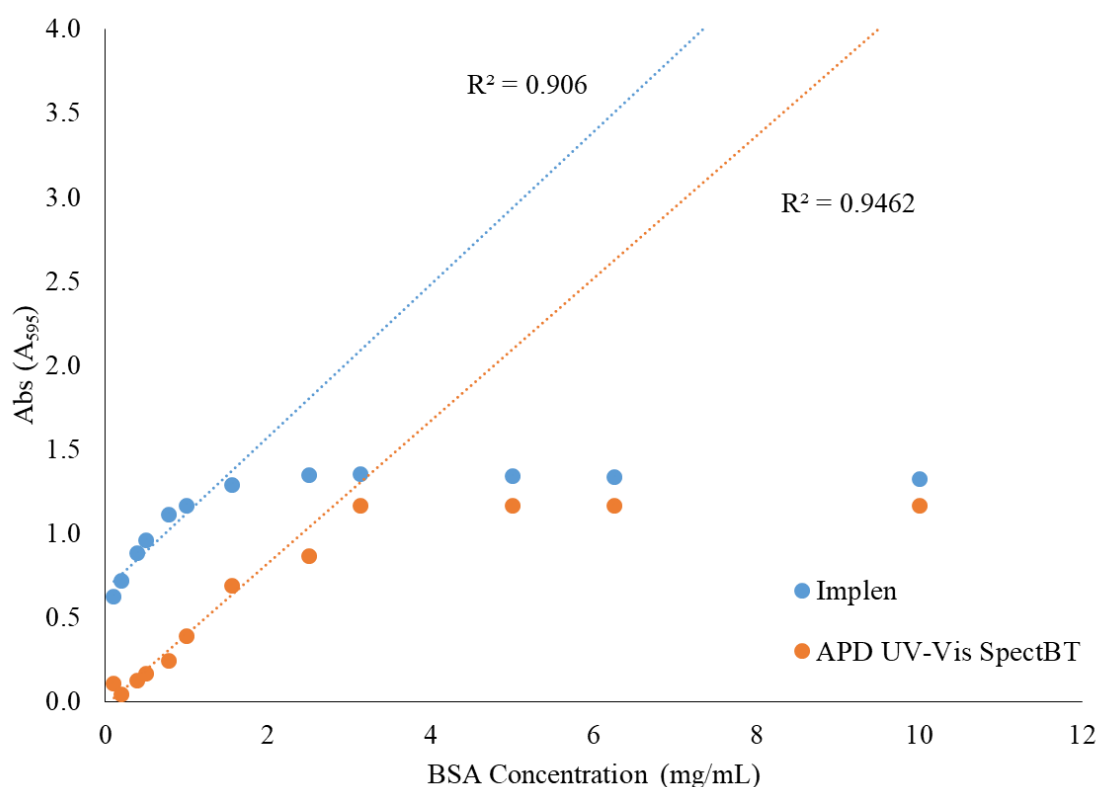


Figure 6. A_{595} standard curves of BSA from the Implen P330 and APD UV-Vis SpectBT

More accurate Abs measurements at lower BSA concentrations were observed for APD UV-Vis SpectBT when compared to protein analyses at 280 nm. It should be noted that the Bradford assay is also more resistant to interference from reducing agents and can be used to detect proteins concentrations as low as 1 µg reliably [32].

3.3. LAMP Analysis at A₅₉₅

The LAMP technology involves a highly-specific amplification of nucleic acid target through the use of primers and provides clear visual colorimetric indications on the amplification product [13]. In the presence of target DNA or RNA, the LAMP solution turns from pink to yellow but remains pink in the absence of target nucleic acid. The result was analysed through a spectrophotometer. Like the Implen P330, APD UV-Vis SpectBT showed more distinct differences in A₅₉₅ readings when measuring the various Lamp tubes (Figure 7A) for better distinct determination of positivity (yellowness). Zero SE was achieved under most conditions showing good repeatability.

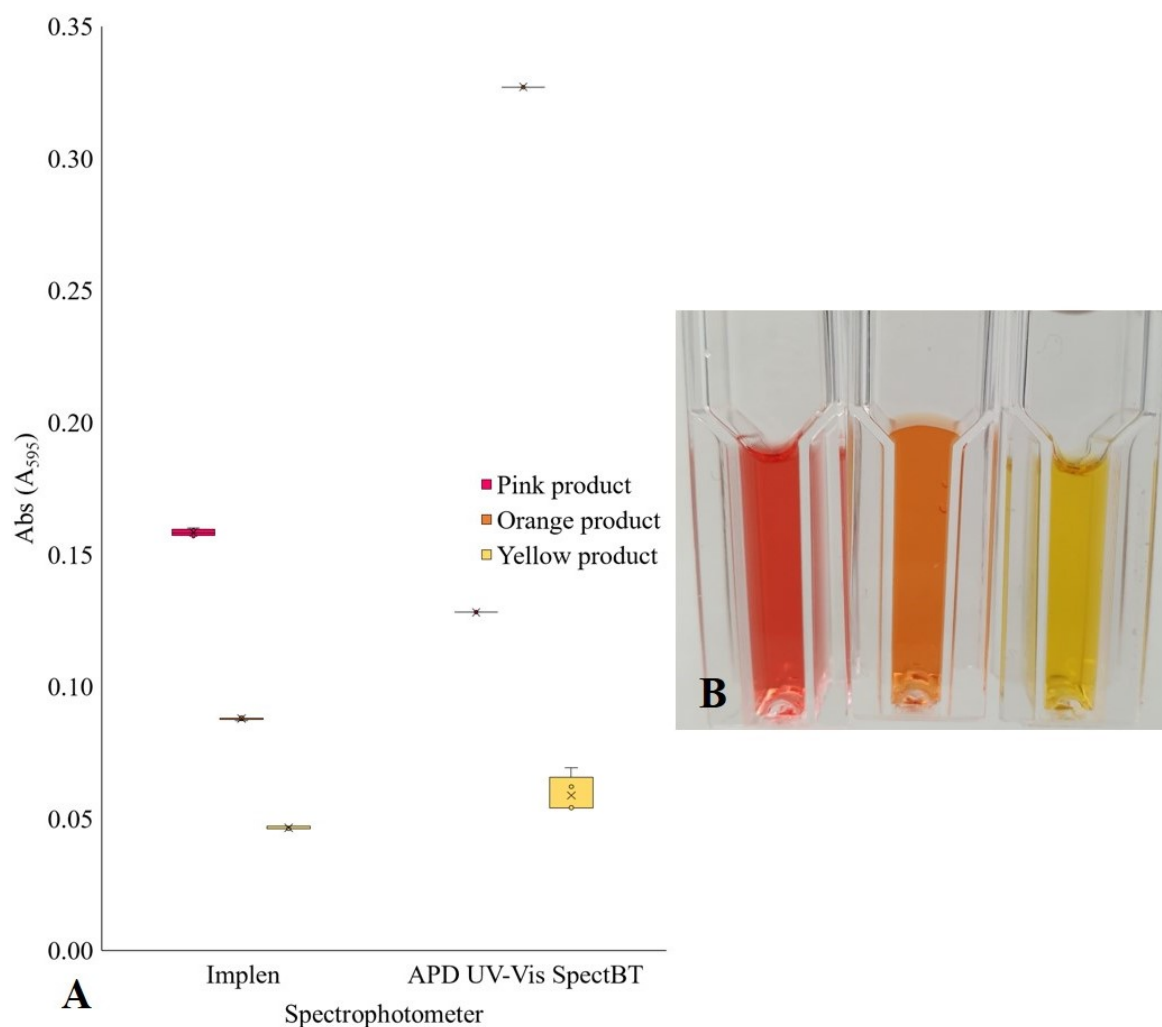


Figure 7. (A) Plot of LAMP at A₅₉₅ for Implen P330 and APD UV-Vis SpectBT and (B) LAMP reagent results

In the example of LAMP detection for COVID-19, the better separation of colorimetric products and portability of this device can help facilitate the administration of rapid LAMP test kits analyses on-the-go.

3.4. Experimental Summary

APD UV-Vis SpectBT is shown to be comparable to a commercial counterpart (Implen P330) in the UV-Vis application tested, with some areas of higher detection limit ~75mg/mL for protein concentrations despite a poorer fit in protein measurements at 280 nm. Our device performed well with the Bradford assay reagent at 595 nm.

APD UV-Vis SpectBT achieved lower sensitivity levels for A₂₈₀ protein measurements compared to the Implen P330. Future enhancements to its sensitivity

performance can be done by improving the Arduino firmware algorithms for the analog-to-digital conversion phase or using more sensitive sensors.

Besides measuring the Abs of proteins, APD UV-Vis SpectBT shows potential in measuring the Abs of DNA. Measurements of the UV absorbance for both APD UV-Vis SpectBT and Implen P330 were measured and compared at 280 nm (A_{280}) against plasmid DNA of concentrations of 10 - 500 ng/ μ L (Supplementary Figure 8), with the APD UV-Vis SpectBT obtaining a wider measuring range but lower R^2 value (Supplementary Figure 8). As DNA absorbance readings are commonly performed at 260 nm (A_{260}) [29], our device's 265 – 310 nm UV LED can be swapped with a 260 nm UV LED for optimal measurement of DNA Abs.

4. Conclusion

UV-Vis spectrophotometry is a commonly employed analytical technique. To improve its deployment, we describe the development of a DIY spectrophotometer using low-cost and readily available electronic components. Not only is this an economical solution for financially challenging circumstances, it also provides portability to support on-the-go analyses.

Our developed APD UV-Vis SpectBT spectrophotometer and APD SpectBT Pro Android mobile app provides seamless data acquisition, transfer and storage to a smartphone device. Experimental validation against commercial counterpart (Implen P330) showed good performance in measuring Abs of proteins at 595nm (A_{595}) wavelength and repeatability alongside higher differentiation for LAMP reactions. However, measurements at 280nm (A_{280}) wavelength fared slightly poorer. The use of higher quality electronic components may help to improve the goodness-of-fit and sensitivity of device. Nonetheless, our self-assembled APD UV-Vis SpectBT has high validity to measure Abs of biomolecules in selected wavelengths of light.

Author Contributions: P.-J.J. drafted the proposal, executed the statistical analysis, interpreted the results, and revised the manuscript. N.W.-J.G interpreted some results and revised the manuscript. S.M-X.T designed the 3D printing casing. S.K.-E.G. reviewed the proposal, performed some experiments, interpreted some results, and supervised the whole project. All authors have read and approved the final manuscript.

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